Voices of Graduation: A Comprehensive NLP Analysis of Commencement Speeches

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By

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Certificate

This is to certify that work described in this thesis entitled **"Voices of Graduation: A Comprehensive NLP Analysis of Commencement Speeches"** has been carried out by Tanushri Shetty under my supervision. I certify that this is his/her bonafide work. The work described is original and has not been submitted for any degree to this or any other University.

Date:

Place:

____
(Supervisor)

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This is to submit that this written submission in my report entitled **"Voices of Graduation: A Comprehensive NLP Analysis of Commencement Speeches"** (Semester IV) represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I stuck to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/ data/ fact/ source in my submission. I understand that any violation of the above will be cause for disciplinary action by the School and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Contents

Abstract

In this project, we conducted a comprehensive analysis of 322 commencement speeches delivered across various universities in the United States. We created our dataset to include key attributes such as speaker information, university details, speech year, transcript content, speaker profession, and school location. Utilizing natural language processing (NLP) techniques, we extracted insights from the textual data.

First, we transformed the transcripts into a term frequency-inverse document frequency (TF-IDF) matrix, focusing on nouns. We then applied non-negative matrix factorization (NMF) to identify seven predominant topics addressed in the speeches. We analyzed the trends of these topics over the years and explored variations based on the professions of the speakers and the location of the universities.

Next, we employed VADER sentiment analysis to evaluate the polarity of the speeches over time. Additionally, we examined sentiment trends based on speaker professions. Subsequently, we investigated the relationship between polarity and subjectivity using TextBlob, visualizing the results through a scatter plot. Furthermore, we leveraged the NRC emotion lexicon to identify and analyze the most common emotions expressed in the speeches.

Our findings offer valuable insights into the thematic content, sentiment dynamics, and emotional undercurrents of commencement addresses, shedding light on the nuanced aspects of this influential genre of public discourse.

Introduction

Commencement speeches offer a unique window into the aspirations and values instilled in graduating students. This project delves into the thematic content and emotional undercurrents of commencement speeches delivered at various universities across the United States. We analyze 322 speeches, focusing on the language used and the sentiment expressed. By employing Natural Language Processing (NLP) techniques, we aim to uncover the prevailing topics addressed by speakers, how these topics have evolved over time, and any potential variations based on the speaker's profession or the university's location. Additionally, we explore the sentiment of the speeches throughout their duration and its connection to the overall message conveyed.

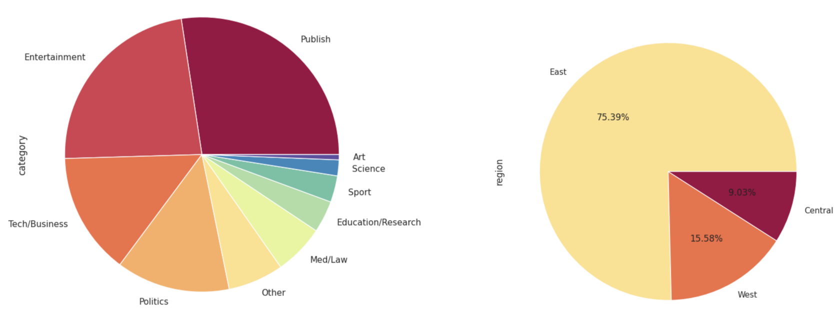
Data Description

The data comprises transcripts from 322 graduation commencement speeches delivered by renowned personalities across various universities in the United States.

The dataset is constructed from transcripts. Each entry in the dataset contains essential information such as the name of the speaker, the year of the speech, the transcript of the speech, and the hosting university.

To enrich the dataset and provide additional context, supplementary details including columns specifying the profession of each speaker and the geographical location of the hosting university are added to the dataset. The addition of this auxiliary information facilitate a more comprehensive and interesting analysis of the speeches and their underlying themes.

The below figure provides the distribution of the professions of the speakers and the location of the universities in the dataset.



(a) Profession (b) Location

Figure 1: Distribution of Parameters

Text Preprocessing

In this project we are going to be using the dataset to analyse the common topics emphasised in these speeches. To begin we will have to do some preprocessing to the text.

Document Term Matrix

A document-term matrix (DTM) is a tabular representation of a text corpus.

Rows: Each row in the matrix represents a document from the corpus. This could be a single text document, a webpage, an article, or any unit of text that is being analyzed.

Columns: Each column represents a unique term (word or phrase) present in the entire corpus. The columns typically exclude common stop words (e.g., "the", "and", "is") and may involve techniques like stemming or lemmatization to normalize the terms.

Cell Values: The cells of the matrix contain numerical values representing the frequency, weight, or some other measure of the occurrence of the corresponding term in the respective document. Common representations include raw term frequencies, term frequency-inverse document frequency (TF-IDF) scores, or binary indicators (1 if the term appears in the document, 0 otherwise).

DTMs are widely used in NLP and text mining tasks for various purposes. In this we use the binary and TF-IDF representation of DTM are used as input data for algorithms to identify topics within the corpus.

Lemmatization

Text lemmatization is a NLP technique used to reduce words to their base or canonical form, known as the lemma. The lemma is the dictionary form of a word, which represents its canonical meaning regardless of its inflected or derived forms. The process of lemmatization involves identifying the lemma for each word in a text based on its intended meaning and context. This process ensures that words with similar meanings are grouped together, which is crucial for accurately identifying topics within a corpus.

Here's why text lemmatization is necessary for topic modeling:

Reducing Vocabulary Size:Lemmatization reduces the total number of unique words in the corpus by converting inflected or derived forms of words to their base forms. This helps in simplifying the vocabulary, making it easier for topic modeling algorithms to identify meaningful patterns and clusters.

Improving Topic Coherence:By standardizing words to their lemmas, lemmatization helps in aggregating words with similar semantic meanings. This improves the coherence of topics generated by the topic modeling algorithm, as words with similar meanings are more likely to be grouped together within the same topic.

Handling Synonyms and Variants:Lemmatization helps in handling synonyms and word variants effectively. By converting words to their base forms, lemmatization ensures that synonyms are treated as the same word, which prevents topic modeling algorithms from generating redundant or overlapping topics based on variations of the same word.

Overall, text lemmatization plays a crucial role in improving the accuracy, coherence, and interpretability of topics generated by topic modeling algorithms. It helps in standardizing the vocabulary, aggregating words with similar meanings, and ensuring that topics are representative of the underlying themes present in the corpus.

In this project we use the Python library NLTK to create a new dataframe where column containing the transcript column has been lemmatized which is later used in topic modelling. Then using the lemmatized transcript we

can create a new document term matrix and find out the most common words used in the speech after removing stop words and some other commonly used words.

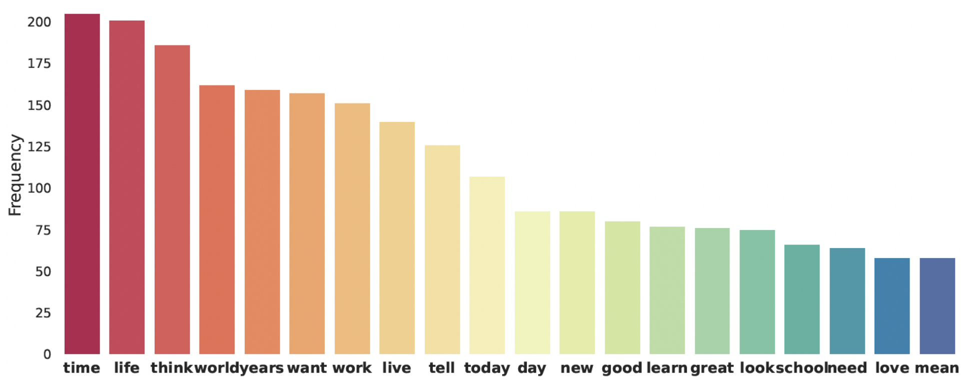


Figure 2: Bar Graph of Most Common Words used in Speeches

Topic Modelling

Topic modeling serves as a powerful method to unveil latent semantic structures embedded within a text corpus, enabling the automatic extraction of underlying topics within it. Essentially, it entails employing statistical modeling techniques, often within the realm of unsupervised machine learning, to scrutinize textual data and discern clusters or sets of words that exhibit similarity or coherence.

Topic modeling operates by discerning latent themes within a large collection of text, wherein documents pertaining to a particular theme tend to exhibit certain words with higher frequency. This process involves analyzing the distribution of words across documents and grouping together words that frequently co-occur. In essence, topic modeling extracts underlying patterns in text data by identifying clusters of words that tend to appear together, thus enabling the creation of coherent topic clusters that represent the main themes present in the corpus.

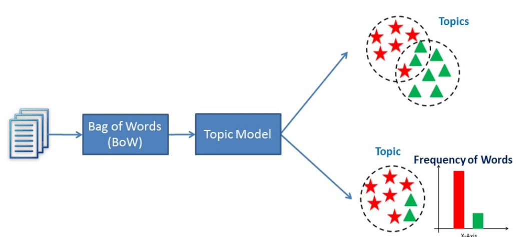
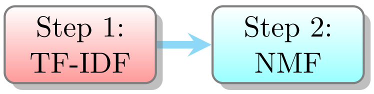


Figure 3: A visualisation of how topic modelling works

Method Used



Step 1: TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF is a statistical measure commonly used in NLP to evaluate the importance of a word in a document relative to a collection of documents (corpus). It quantifies how relevant a word is to a document within a corpus by considering both its frequency within the document (TF) and its rarity across the entire corpus (IDF).

Here's how TF-IDF works:

1. **Term Frequency (TF)**: This component measures the frequency of a term (word) within a document. It's calculated as the number of times a term appears in a document divided by the total number of terms in the document. Essentially, it represents how often a term occurs within the document.

   TF(t, d) = (Number of times term t appears in document d) / (Total number of terms in document d)

2. **Inverse Document Frequency (IDF)**: This component measures the rarity of a term across the entire corpus. It's calculated as the logarithm of the total number of documents in the corpus divided by the number of documents containing the term, with the result inverted. Essentially, it penalizes terms that appear in many documents and rewards terms that appear in few documents.

   IDF(t) = $\log_e$(Total number of documents / Number of documents containing term t)

3. **TF-IDF Score**: The TF-IDF score for a term in a document is computed by multiplying its TF value by its IDF value. This results in a score that reflects both the importance of the term within the document and its rarity across the corpus.

   TF-IDF(t, d) = TF(t, d) * IDF(t)

By calculating TF-IDF scores for each term in each document, you obtain a matrix known as the TF-IDF matrix. High TF-IDF scores indicate terms that are both frequent within the document and rare across the corpus, making them potentially significant in representing the content and themes of the document.

Step 2: Non-negative Matrix Factorization (NMF) Modelling

NMF is a dimensionality reduction technique that is commonly used for topic modeling in NLP. NMF directly factorizes the term-document matrix into two non-negative matrices: one representing the topics and the other representing the document-topic distributions.

Here's how NMF topic modeling works:

1. **Term-Document Matrix**: The input to NMF is a term-document matrix, where each row corresponds to a term (word) and each column corresponds to a document. The cells of the matrix contain the frequency of each term in each document or some other measure like TF-IDF..

2. **Non-negative Matrix Factorization**: NMF decomposes the term-document matrix into two non-negative matrices: W (the basis matrix) and H (the coefficient matrix). These matrices are calculated iteratively to minimize the reconstruction error between the original matrix and the product of W and H.

    - W (Topics): The W matrix represents the topics, where each column corresponds to a topic, and each row corresponds to a term. The values in W represent the importance of each term within each topic.
    - H (Document-Topic Matrix): The H matrix represents the document-topic distribution, where each row corresponds to a document, and each column corresponds to a topic. The values in H represent the contribution of each topic to each document.

3. **Interpretation**: Once the matrices W and H are computed, the resulting topics can be interpreted by examining the terms with the highest weights in each topic. Similarly, the document-topic distribution provides insights into the distribution of topics across the documents in the corpus.

4. **Number of Topics**: The number of topics in NMF topic modeling is a hyperparameter that needs to be specified beforehand. Researchers typically experiment with different numbers of topics to find the optimal balance between granularity and interpretability.

NMF has several advantages for topic modeling, including simplicity, interpretability, and computational efficiency. However, it also has limitations, such as the need to specify the number of topics in advance and the fact that it may not capture correlations between terms as effectively as probabilistic models like LDA. Overall, NMF is a versatile technique for discovering latent topics within a corpus of text data.

In this project we have followed the following process:

1. **Text Preparation**: The commencement speeches are first preprocessed, focusing specifically on the text element noun, using techniques like tokenization and filtering. This step aims to clean and prepare the text data for analysis.

2. **Tf-idf Vectorization**: The text data consisting of nouns is then transformed into a numerical representation using the Term Frequency-Inverse Document Frequency (Tf-idf) vectorization technique.

3. **NMF Topic Modeling**: NMF is applied to the Tf-idf matrix to decompose it into two matrices: one representing topics and the other representing the distribution of topics in each document. Seven Topics are extracted in this project

4. **Topic Assignment**: Each document is assigned to the topic with the highest weight based on the NMF output.

5. **Top Words per Topic**: For each identified topic, the top words contributing to that topic are extracted. These words represent the key terms associated with each topic and provide interpretability to the generated topics.

6. **Output**: The final output is a DataFrame containing the identified topics along with the top words associated with each topic. This allows for easy interpretation and understanding of the main themes present in the commencement speeches. We then add new column 'topic' to the dataframe containing the details of the speaker, speech and university which contains the predominant topic for the respective speech.

The final topics we get after applying this process to our dataset are:

- **Family, Responsibility & Advice** characterised by words like parents, kids, advice and job.
- **Nation, Rights & Freedom** characterised by words like government, nation, freedom and rights.
- **Women's Voice** characterised by words like women, girls and work.
- **Education & University** characterised by words like university, education, school and college.
- **Ambition** characterised by words like passion, success, dreams and business.
- **Arts, Science & Knowledge** characterised by words like arts, science, knowledge and culture.
- **Emotions** characterised by words like fear, love, faith and death.

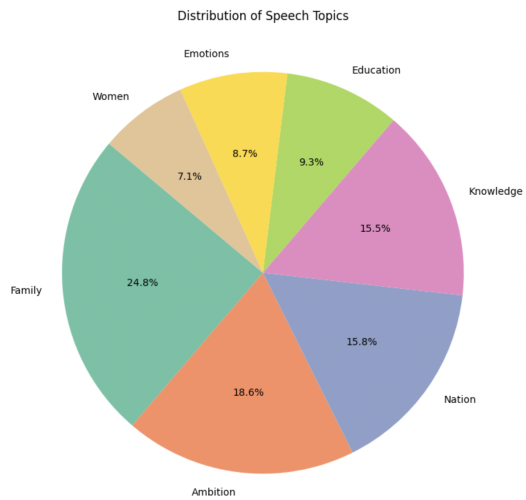


Figure 4: Distribution of Topics in the Dataset

In figure 4, we can observe that Family, Responsibility & Advice; Ambition and Nation, Rights & Freedom in that order are the three most widely talked about topics in the speeches of the dataset.

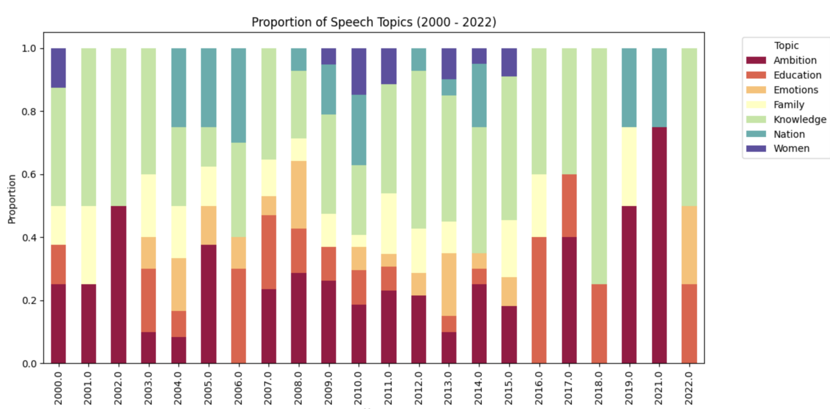


Figure 5: Proportion of Topics of Speeches 2000-2022

In figure 5, we can observe that speakers consistently talk about Arts, Science & Knowledge in recent years. Another topic regarding which high proportion of speeches are given in recent years is Ambition. We also see a slight spike in speeches regarding women's voices in recent years which can be attributed to more awareness regarding the topic in recent years.

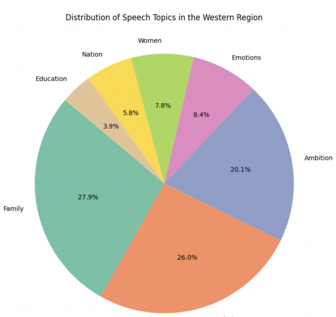


Figure 6: Distribution of Topics West

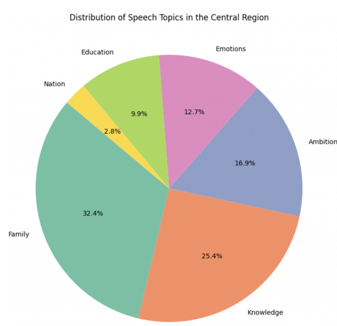


Figure 7: Distribution of Topics Central

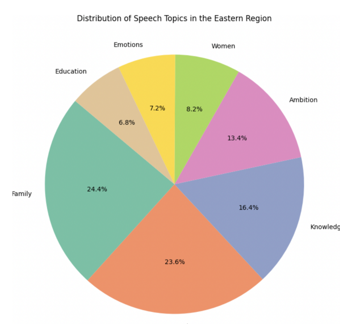
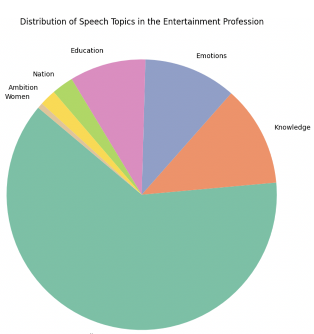
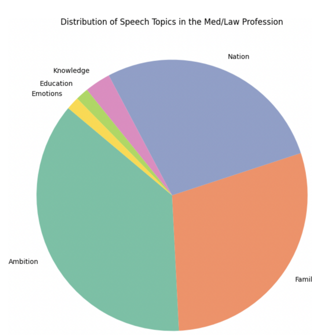
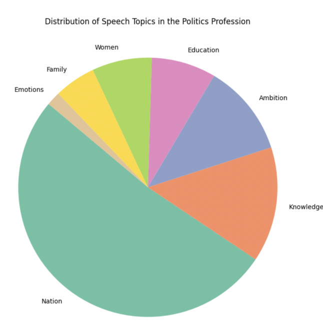


Figure 8: Distribution of Topics East

In figures 6,7 and 8 we can observe the distribution of topics with respect to the regions they were given in. An interesting observation in this distribution is that in the central region which is considered a more conservative region in America none of the speeches were around the topic of women's voice.







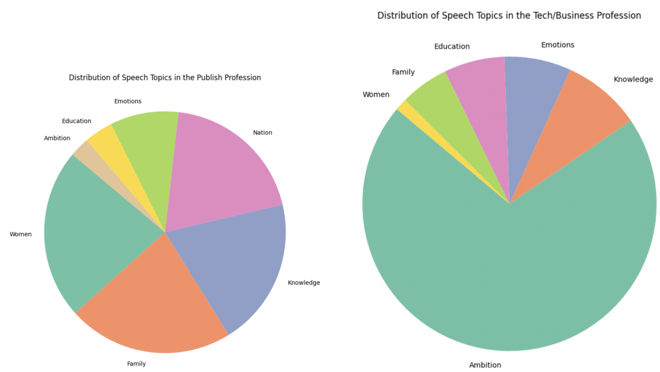


Figure 9: Distribution of Topics by Speaker Profession

In figure 9 (refer both page 17 & 18) we can observe the trend of topics according to the profession of speakers. We observe

- Family, Responsibility & Advice is the most talked about talk among speakers from the entertainment industry
- Family, Responsibility & Advice; Ambition and Nation, Rights & Freedom seem to be the most and almost equally popular topics amongst medicine and law professionals.
- Nation, Rights & Freedom as expected seems to be the most popular topic among politicians.
- Women's Voice; Nation, Rights & Freedom; Family, Responsibility & Advice and Arts, Science & Knowledge are the most spoken about topics by writers and individuals in the publishing industry.
- Ambition is overwhelming the most popular topic touched upon by technology and business professionals.

Sentiment Analysis

The method for sentiment analysis used in this project is VADER. VADER (Valence Aware Dictionary and sentiment Reasoner) is a lexicon and rule-based sentiment analysis tool used to assess the sentiment of text data. VADER is widely used in NLP tasks to quickly and effectively analyze the sentiment of textual content.

Here's how VADER works:

1. **Pre-built Lexicon**: VADER comes with a pre-built lexicon containing thousands of words, along with their associated sentiment scores. Each word in the lexicon is assigned polarity scores representing the positivity, negativity, and neutrality of the word.

2. **Rule-based Analysis**: VADER employs a set of rules and heuristics to analyze the sentiment of text. It takes into account factors such as the presence of capitalization, punctuation, degree modifiers (e.g., "very"), and emoticons to better understand the sentiment expressed in the text.

3. **Scoring**: When analyzing a piece of text, VADER tokenizes the text into individual words and computes a sentiment score for each word based on its polarity score in the lexicon. It then aggregates these scores to generate an overall sentiment score for the entire text.

4. **Compound Score**: The overall sentiment score produced by VADER ranges from -1 (extremely negative) to +1 (extremely positive), with scores closer to zero indicating a more neutral sentiment.

5. **Strengths and Limitations**: VADER is effective for sentiment analysis of short, casual, texts. Its rule-based approach allows for efficient sentiment analysis without the need for extensive training data. However, VADER may not perform as well with longer, more complex text or in domains with specialized language and terminology.

In this project we segment each speech into 10 equal parts and calculate the polarity score for each segment in order to give us an idea about how the polarity changes over the duration of the speech.

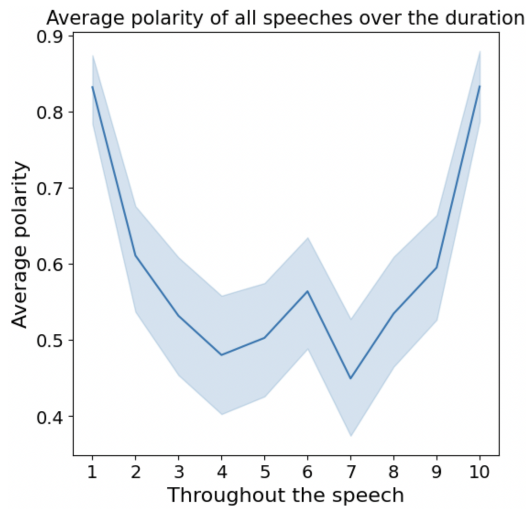


Figure 10: Average Polarity of All Speeches throughout the duration

We can observe in figure 10 that in general speeches have an overall positive sentiment throughout the duration.

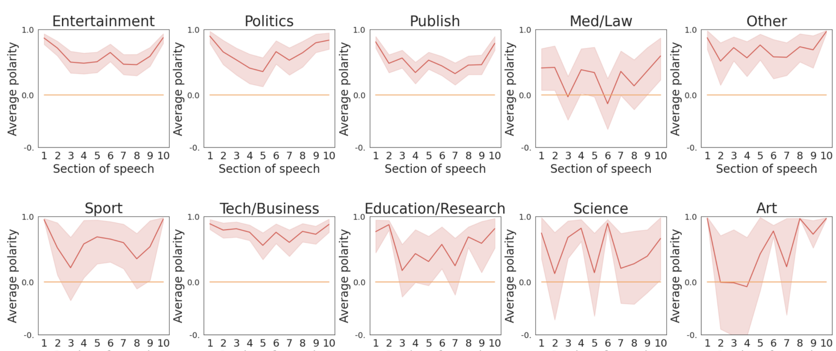


Figure 11: Average Polarity by Profession

In figure 11 we can look at the average polarity grouped by professions. We can observe a general positive sentiment with some variation edging towards negative sentiments only in speeches given by medicine, law and art professionals.

Using the python library textblob we can also find the subjectivity of a text. The subjectivity score ranges from 0 to 1, where 0 indicates complete objectivity (i.e., the text is purely factual) and 1 indicates complete subjectivity (i.e., the text is entirely opinionated).

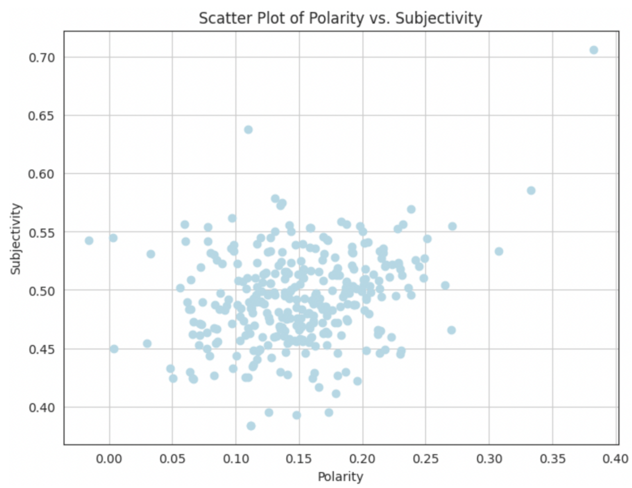


Figure 12: Scatter Plot of Subjectivity vs Polarity

We can observe in figure 12 that in general the subjectivity score of speeches lies around the 0.5 area indicating that the speeches are fairly equally divided between being objective and subjective in nature.

Emotion Analysis

The NRC (National Research Council) Emotion Lexicon is a lexicon that contains a list of words along with their associated emotions. It is widely used in NLP for emotion analysis tasks, allowing one to assess the emotional content of text data.

Here's how we have performed emotion analysis using the NRC Emotion Lexicon:

1. **Load the Emotion Lexicon**: First, we have loaded the NRC Emotion Lexicon into our Python environment as a csv file.
2. **Emotion Analysis**: For each word in the text, check if it exists in the NRC Emotion Lexicon. If it does, retrieve the associated emotions for that word.
3. **Aggregate Emotions**: Aggregate the emotions associated with all words in the text to determine the overall emotional content. We can then count the occurrences of each emotion in a speech.
4. **Normalization**: We the normalize the scores for different emotions for each speech in order to better interpret the data.

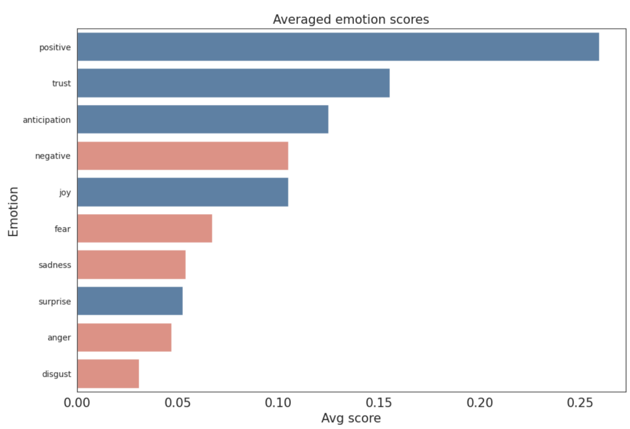


Figure 13: Averaged Emotion Scores for All Speeches

In figure 13 like previously seen in sentiment analysis we can see that positive emotions have a higher scores than negative emotions with positive, trust and anticipation being the top emotions.

Conclusion

This project has demonstrated the effectiveness of NLP in analyzing the thematic content and sentiment of commencement speeches. The findings offer valuable insights into the evolving landscape of these addresses, highlighting the topics most emphasized and the emotional undercurrents that resonate with graduating students.

Limitations

The dataset is relatively small and while conducting analysis on a small dataset can yield valuable insights, it also comes with inherent limitations that can impact the robustness and generalizability of the findings. It is hard to find concrete trends over time especially when it came to topic modelling as the number of speeches in the dataset from each year were less.

Future Work

Future research could explore a larger dataset, incorporating additional NLP techniques to delve deeper into the stylistic choices and impactful language employed within these speeches. It could also explore text generation from a text corpus and usage of topic modelling in text generation.

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